Leccion1MarcelaGuerrero

November 4, 2026

[3]:
```
#Tratamiento de datos
import numpy as np
import pandas as pd

#Gráficos
import matplotlib.pyplot as plt

#Preprocesado y modelado
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from numpy import linalg as LA
```

[7]:
```
df = pd.DataFrame(np.array([[7,6.5,9.2,8.6,8],
                            [7.5,9.4,7.3,7,7],
                            [7.6,9.2,8,8,7.5],
                            [5,6.5,6.5,7,9],
                            [6,6,7.8,8.9,7.3],
                            [7.8,9.6,7.7,8,6.5],
                            [6.3,6.4,8.2,9,7.2],
                            [7.9,9.7,7.5,8,6],
                            [6,6,6.5,5.5,8.7],
                            [6.8,7.2,8.7,9,7]]),
                  index = ['Lucia', 'Pedro', 'Ines','Luis','Andres', 'Ana',␣
 ↪'Carlos', 'Jose', 'Sonia', 'Maria'],
                  columns = ['Matematicas', 'Ciencias', 'Español',␣
 ↪'Historia','EdFisica'])

df
```

[7]:

	Matematicas	Ciencias	Español	Historia	EdFisica
Lucia	7.0	6.5	9.2	8.6	8.0
Pedro	7.5	9.4	7.3	7.0	7.0
Ines	7.6	9.2	8.0	8.0	7.5
Luis	5.0	6.5	6.5	7.0	9.0
Andres	6.0	6.0	7.8	8.9	7.3
Ana	7.8	9.6	7.7	8.0	6.5

```
Carlos          6.3       6.4       8.2       9.0       7.2
Jose            7.9       9.7       7.5       8.0       6.0
Sonia           6.0       6.0       6.5       5.5       8.7
Maria           6.8       7.2       8.7       9.0       7.0
```

[8]:
```
#Determinar el numero de renglones de la base de datos
index = df.index
renglones = len(index)
renglones
```

[8]:
```
10
```

[8]:
```
#Estandarización de las variables
df2 = StandardScaler().fit_transform(df)
df2
```

[8]:
```
array([[ 0.23263076, -0.7529862 ,  1.78848525,  0.65792263,  0.65858084],
       [ 0.78651352,  1.14584856, -0.53899555, -0.84590053, -0.47690337],
       [ 0.89729007,  1.01489444,  0.31849737,  0.09398895,  0.09083874],
       [-1.98290027, -0.7529862 , -1.51898747, -0.84590053,  1.79406505],
       [-0.87513476, -1.0803715 ,  0.07349939,  0.93988948, -0.13625811],
       [ 1.11884317,  1.27680268, -0.0489996 ,  0.09398895, -1.04464547],
       [-0.5428051 , -0.81846326,  0.56349535,  1.03387842, -0.24980653],
       [ 1.22961972,  1.34227974, -0.29399757,  0.09398895, -1.61238758],
       [-0.87513476, -1.0803715 , -1.51898747, -2.25573474,  1.45341979],
       [ 0.01107766, -0.29464677,  1.1759903 ,  1.03387842, -0.47690337]])
```

[9]:
```
#Calculo de la matriz de correlaciones para la matriz transformada
A = (1/ renglones) * np.dot(df2.T,df2)
A
```

[9]:
```
array([[ 1.        ,  0.85407878,  0.38457424,  0.20719425, -0.78716269],
       [ 0.85407878,  1.        , -0.02005218, -0.02153942, -0.68772056],
       [ 0.38457424, -0.02005218,  1.        ,  0.82091619, -0.36554342],
       [ 0.20719425, -0.02153942,  0.82091619,  1.        , -0.50800132],
       [-0.78716269, -0.68772056, -0.36554342, -0.50800132,  1.        ]])
```

[11]:
```
#Entrenamiento del modelo PCA con escalado de los datos
pca_pipe = make_pipeline(StandardScaler(), PCA())
pca_pipe.fit(df)

#Se extrae el modelo entrenado del pipeline
modelo_pca = pca_pipe.named_steps['pca']
print("Eigenvalores:")
results = LA.eigvals(A)
print(results)
```

```
Eigenvalores:
[2.89324967 1.62865042 0.34659605 0.00889139 0.12261246]
```

[12]:
```
# Porcentaje de la varianza explicada por cada nuevo componente
print("Porcentaje de varianza explicada por componente")
print(modelo_pca.explained_variance_ratio_)
```

```
Porcentaje de varianza explicada por componente
[0.57864993 0.32573008 0.06931921 0.02452249 0.00177828]
```

[14]:
```
# Cálculo de eigenvectores
print("Eigenvectores (por renglón):")
pd.DataFrame(data = modelo_pca.components_,
             columns = df.columns,
             index = ['PC1', 'PC2', 'PC3','PC4','PC5'])
```

```
Eigenvectores (por renglón):
```

[14]:

	Matematicas	Ciencias	Español	Historia	EdFisica
PC1	-0.526644	-0.424936	-0.359147	-0.352697	0.537302
PC2	-0.270496	-0.508072	0.562082	0.586490	0.093746
PC3	0.438201	0.040495	0.562276	-0.394180	0.578626
PC4	-0.261218	0.673627	-0.070086	0.446645	0.523056
PC5	-0.623878	0.325390	0.483747	-0.420433	-0.306794

[15]:
```
#Proyecciones de los componentes
proyecciones = np.dot(modelo_pca.components_, df2.T)
proyecciones = pd.DataFrame(proyecciones, index = ['PC1', 'PC2',␣
 ↪'PC3','PC4','PC5'])
proyecciones = proyecciones.transpose().set_index(df.index)
proyecciones
```

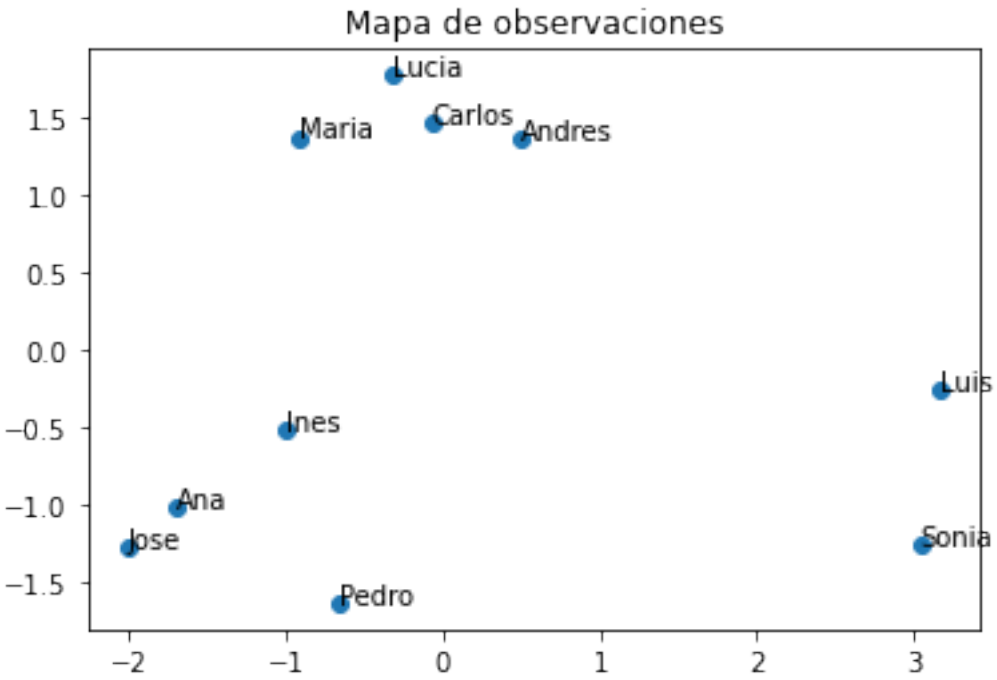
[15]:

	PC1	PC2	PC3	PC4	PC5
Lucia	-0.323063	1.772525	1.198801	-0.055015	-0.003633
Pedro	-0.665441	-1.638702	0.145476	-0.023065	0.123377
Ines	-1.002547	-0.515692	0.628888	0.516444	-0.142876
Luis	3.172095	-0.262782	-0.381960	0.677777	0.062504
Andres	0.488868	1.365402	-0.835236	-0.155792	-0.123367
Ana	-1.708633	-1.021700	-0.127077	0.066833	-0.025292
Carlos	-0.067586	1.462336	-0.506240	-0.117928	-0.013124
Jose	-2.011855	-1.275865	-0.542150	-0.197787	-0.017434
Sonia	3.042030	-1.254881	0.448829	-0.639999	-0.037885
Maria	-0.923869	1.369359	-0.029330	-0.071467	0.177730

[18]:
```
x = proyecciones.iloc[:,0]
y = proyecciones.iloc[:,1]
z = df.index
x = x.to_numpy()
```

```
y = y.to_numpy()
fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



[19]:
```
componentes_2 = pd.DataFrame(data= modelo_pca.components_,
                             columns = df.columns,
                             index = ['PC1', 'PC2', 'PC3','PC4','PC5'])
componentes_2 = componentes_2.iloc[0:2,:]
componentes_2 = componentes_2.T
componentes_2
```

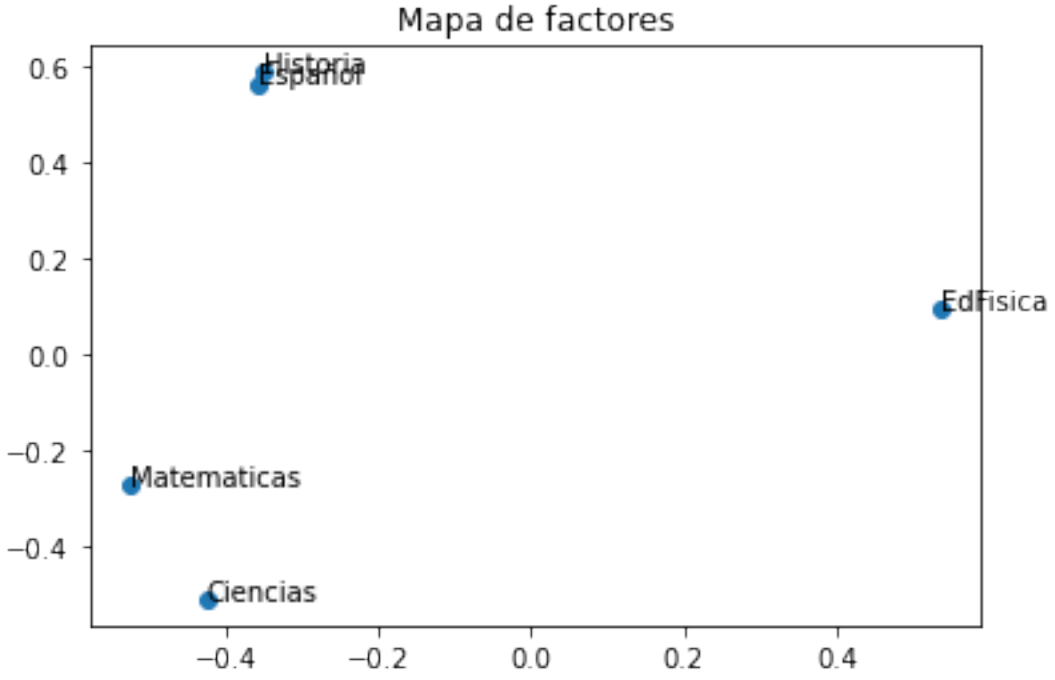
[19]:

	PC1	PC2
Matematicas	-0.526644	-0.270496
Ciencias	-0.424936	-0.508072
Español	-0.359147	0.562082
Historia	-0.352697	0.586490
EdFisica	0.537302	0.093746

[20]:
```
x = componentes_2.iloc[:,0]
y = componentes_2.iloc[:,1]
```

```
z = componentes_2.index
x = x.to_numpy()
y = y.to_numpy()
fig, ax = plt.subplots()
ax.set_title("Mapa de factores")
ax.scatter(x,y)

for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



[21]:
```
#Procedimiento para obtener la matriz estandarizada original
original = np.dot(modelo_pca.components_.T, proyecciones.T)
original = original.T
print(original)
```

```
[[ 0.23263076 -0.7529862   1.78848525  0.65792263  0.65858084]
 [ 0.78651352  1.14584856 -0.53899555 -0.84590053 -0.47690337]
 [ 0.89729007  1.01489444  0.31849737  0.09398895  0.09083874]
 [-1.98290027 -0.7529862  -1.51898747 -0.84590053  1.79406505]
 [-0.87513476 -1.0803715   0.07349939  0.93988948 -0.13625811]
 [ 1.11884317  1.27680268 -0.0489996   0.09398895 -1.04464547]
 [-0.5428051  -0.81846326  0.56349535  1.03387842 -0.24980653]
 [ 1.22961972  1.34227974 -0.29399757  0.09398895 -1.61238758]
 [-0.87513476 -1.0803715  -1.51898747 -2.25573474  1.45341979]
 [ 0.01107766 -0.29464677  1.1759903   1.03387842 -0.47690337]]
```

[ ]: